

Time: 3 hours

Figures in the right-hand margin indicate marks for individual questions

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- Retrieve the Dependent-Name, Emp-Name, and Department-Name of all dependents.
- Retrieve Emp-Name and Address of employees who don't have any dependent in descending order of Emp-Name.

- c) Create a view containing Dno, Department-Name, No. of employees in the department and Total salary of the employees in the department

3. Answer the following questions in one or two sentences:

2×5 = 10

- (a) Distinguish between relation schema and relation instances.
 (b) What is the atomicity property of transactions?
 (c) What are the different forms of authorization usually provided by DBMS to change the database schema?
 (d) What is the closure of a set of attributes?
 (e) Considering only the number of block transfer and number of seeks, estimate the cost of searching a record using an equality condition on a key field with a primary index.

4. Answer the following questions in short:

3×4 = 12

- (a) When is a set of functional dependencies called minimal?
 (b) When are two schedules called conflict equivalent?
 (c) What are the different statistics about indices that are used for query cost estimation?
 (d) Show that both wound-wait and wait-die scheme of deadlock prevention avoid starvation.

5. Consider a relation schema $R(A, B, C, D, E, F)$ with the set of FDs $F = \{A \rightarrow BC, D \rightarrow E, AD \rightarrow F\}$.

Determine whether the decomposition of R into $R_1(A, B, C)$, $R_2(D, E)$ and $R_3(A, D, F)$ is lossless or not.

5

OR

Find out the key(s) and the highest normal form of R . Normalize R into two higher level normal forms (i.e. if R is in n -th normal form, normalize it to $(n+1)$ th and $(n+2)$ th normal forms.

2+3

6. What is heuristic query optimization what is its advantage over cost-based optimization? State some heuristics rules with examples used in heuristic query optimization.

1+1+3

7. (a) Describe the structure of internal nodes and leaf nodes B+ Tree. 2
 (b) Write an algorithm in pseudocode to search a value in a B+ tree. 4
 (c) State the condition under which the size of the bucket-address table of extendable hashing is increased. 2
 8. (a) How does checkpointing improves the performance of the recovery process? 2
 (b) Write down the actions taken during the checkpointing process in proper sequence. 3
 9. Briefly describe four primary features NoSQL databases. 4

OR

Non-relational
database
Model

(b) Describe Key-value store and document store data models used in NoSQL databases

4

SELECT SALARY
FROM EMPLOYEE
WHERE (SELECT DNO, DEPARTMENT-NAME, EMPNO,
FROM DEPARTMENT, EMPLOYEE)

-----END-----

attribute
alternative

$R_1(A, B, C)$
 $[A^+ = ABC]$

$R_2(D, E)$

$D \rightarrow E$

$R_3(A, D, F)$
 $A \rightarrow BC$
 $AD \rightarrow F$
 $AD = (ADF)$

Mid-term Examination, Autumn 2021
Department of Computer Science and Engineering
CO312 Database System

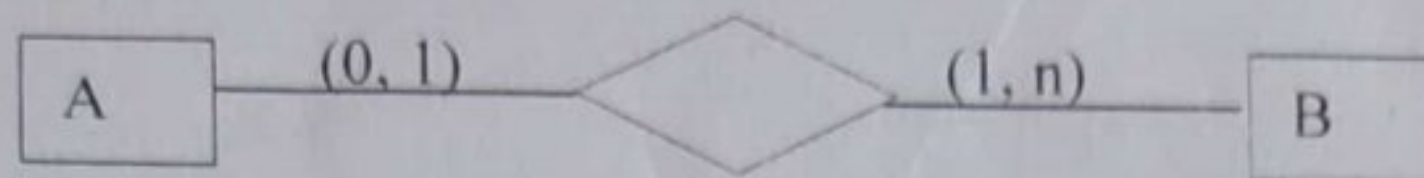
Time: 45 minutes

Total Marks: 20

1. Select the correct answers from the option given for each question:

4 × 1 = 4

a. Consider the relationship shown in the following ER diagram between two entity types:



What are the participation constraint for A and B in the above relationship?

- i) A total, B partial ii) A partial, B total iii) A partial, B partial iv) A total, B total

b. What is the cardinality ratio of the relationship between A and B in the above question

- i) one-to-one ii) many-to-one iii) one-to-many iv) many-to-many

c. Deleting a tuple from a relation may violate

- i) Key and referential integrity constraint iii) Only key constraint
 ii) Only referential integrity constraint iv) All constraints

d. The term **Relation** in relational model refers to:

- i) The relationship between two entities ii) The relational schema
 iii) Set of tuples of a relational schema iv) The relationship with primary key and foreign key

2. State True or False

4 × 1 = 4

- a. Relational algebra is a procedural language.
 b. The set of all attributes of relation schema is a Superkey.
 c. Foreign keys in a relation are not allowed to have NULL values. F
 d. MINUS operations (set difference) is commutative.

3. Consider the two relation schemas:

2 + 2 + 2 = 6

Department(dno, dname) and Employee(eno, name, address, bdate, dept) where 'dept' of Employee is a foreign key and refers to the Department table. Write relational algebra **OR** relational calculus expressions for each of the following queries:

- a) Retrieve eno, name and dname of all employees.
 b) Retrieve eno, name, address of all employees of the 'Administration' department.
 c) Retrieve eno, name, address of all employees of 'Administration' and 'Finance' department.

4. Consider the following description of a miniworld and develop an ER/EER model for it.

6

Students of an institution are allotted rooms of hostels for a period of time. A student may be allotted a different room for different period of time. The rooms allotted may be from different hostels. The common reports needed are: List of students from a hostel and The rooms where a student stayed.

Make any additional assumptions if necessary.

